

ATAR Testbeam Shared Trigger Prototype DAQ

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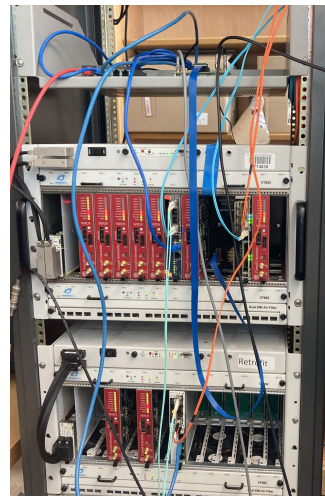
Current Teststand Setup

- The current hardware we have available in our UKy lab:
 - 1 HDSocv1 digitizer
 - 2 uTCA crates
 - ~10 working Cornell WFD5s
 - 1 FC7
 - 2 AMC13s
- Provided a shared external trigger to all front end electronics



HDSocv1

+



g-2 style teststand

Run 244

Running

StopPause

Start: Thu Mar 12 16:27:08 2026

Alarms: On

runStatusSequencer

Data dir: /home/pioneer/data/gm2daq_data/

1773347234 16:27:14.798 2026/03/12 [mhttpd,INFO] Run #244 started

Equipment

Equipment +	Status	Events	Events[/s]	Data[MB/s]
MasterGM2	MasterGM2@dhcp-10-163-105-238	2637	99.6	0.007
AMC13001	AMC13001@dhcp-10-163-105-238	2394	99.6	3.475
AMC13002	AMC13002@dhcp-10-163-105-238	2645	99.6	2.600
EB	EBuilder@dhcp-10-163-105-238	2613	99.9	5.959
HDSoc-00	HDSoc00@::ffff:192.168.1.104	2992	99.6	2.056

Logging Channels

Channel	Events	MB written	Compr.	Disk Level
#0: run00214.mid.lz4	0	0.000	0.0%	2.7%
Lazy Label	Progress	File Name	# Files	Total

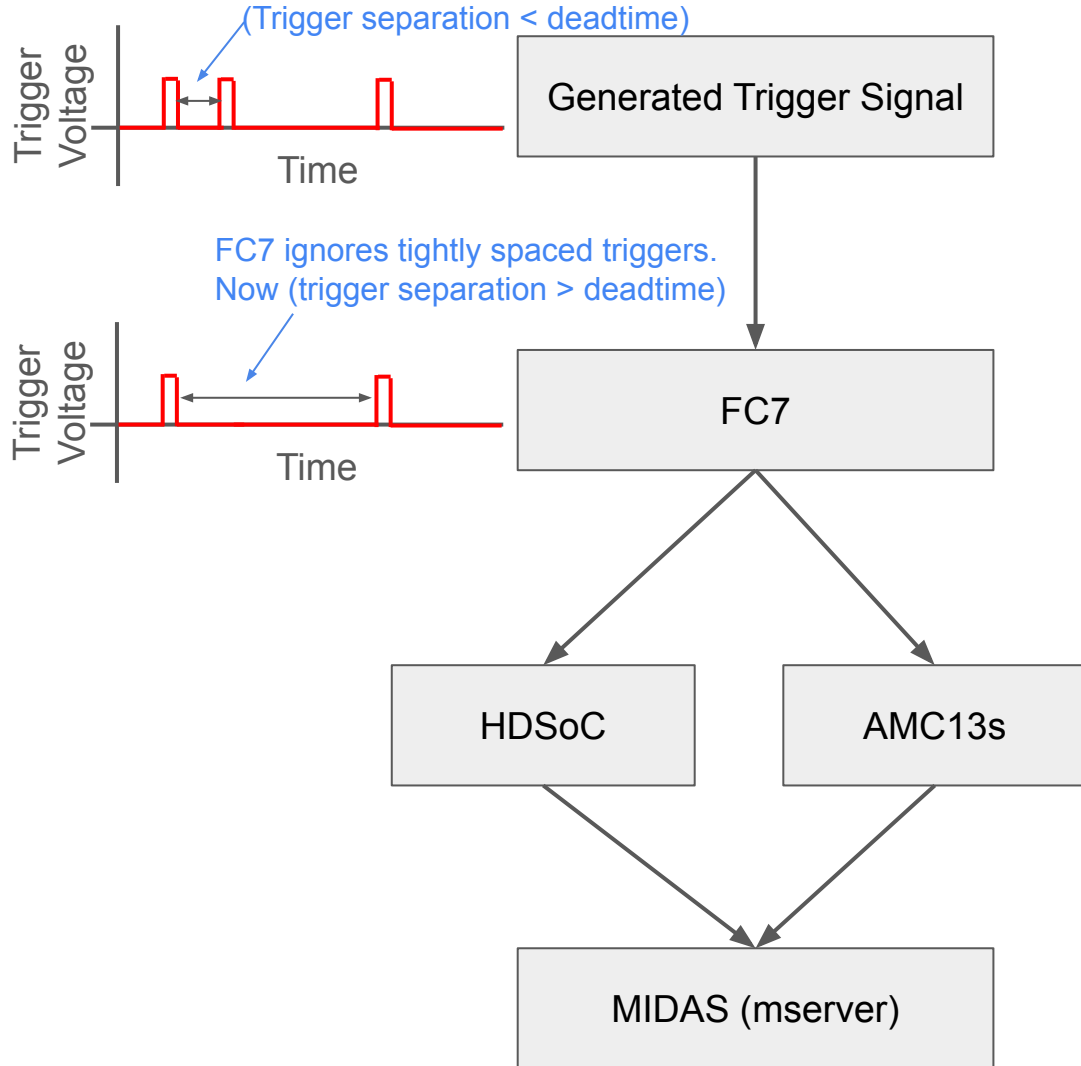
Clients

mserver [dhcp-10-163-105-238]	mhttpd [dhcp-10-163-105-238]	Logger [dhcp-10-163-105-238]
HDSoc00 [::ffff:192.168.1.104]	MasterGM2 [dhcp-10-163-105-238]	AMC13002 [dhcp-10-163-105-238]
AMC13001 [dhcp-10-163-105-238]	Ebuilder [dhcp-10-163-105-238]	

Midas frontends creating data from shared trigger

Shared Trigger Scheme

- Each frontend electronic takes a shared external trigger
 - HDSocv1 frontend 0
 - AMC13 frontend 0
 - AMC13 frontend 1
- Trigger passes through FC7
 - Ungates signal until run start
 - Gates for programmable “deadtime” during runs
 - Idea: Set this high enough such that all frontends are ready for every event
 - Regates signal after run end
- This scheme will only work with a global external trigger



TODO

- Provide more realistic “random” trigger
 - Using a CAEN function generator to generate periodic pulses with random time separation
- Incorporate frontend simulators for SAMPic and MuPix
 - Don't have access to real hardware in our lab, but have simulator frontends available
 - SAMPic should (?) follow a similar shared trigger scheme as the HDSoc
 - Need to better understand MuPix trigger scheme to best design integration
- Put HDSoc on a shared clock
 - Should be possible, need guidance from Nalu/others
- ~~● HDSoc v2/v3 integration (?)~~
 - ~~○ Likely not applicable to ATAR testbeam~~
- Incorporate new systems into DQM framework
 - SAMPic and HDSoc unpackers exist
 - HDSoc has plotting plugins for the webpage already
 - SAMPic does not have plotting plugins written, but this is a short exercise
 - Need to better understand MuPix data output to best design integration

Auxiliary Slides